

Positivity of the computable real number a_3

Executive summary:

- The code defines a sequence $a_3(n) = 1 + 2^{-n} - 72^{-n}$ for $n = 0, 1, 2, \dots$. The computable real a_3 is the limit of this sequence.
- Since $0 < 2^{-n} < 1$ and $0 < 72^{-n} < 1$, both terms 2^{-n} and 72^{-n} decrease monotonically to 0 as $n \rightarrow \infty$. Hence $\lim_{n \rightarrow \infty} (2^{-n}) = 0$ and $\lim_{n \rightarrow \infty} (72^{-n}) = 0$.
- For $n = 0$, $a_3(0) = 1 + 1 - 1 = 1$. For $n \geq 1$, one checks $2^{-n} > 72^{-n}$ (since $36^n > 1$), so $a_3(n) = 1 + 2^{-n} - 72^{-n} > 1$. Thus each term $a_3(n) \geq 1$ (with strict inequality for $n \geq 1$).
- It follows that $a_3 = \lim_{n \rightarrow \infty} a_3(n) = 1 + 0 - 0 = 1$. In particular $a_3 = 1 > 0$. We therefore **prove rigorously that $a_3 > 0$** .
- A table of values for $n = 0, \dots, 6$ confirms $a_3(n) \geq 1.000$ (starting at 1.486 at $n = 1$ and decreasing toward 1). (See table below.)
- In general, **deciding the sign of an arbitrary computable real from its code is undecidable**¹. That is, there is no uniform algorithm to determine positivity for *all* computable reals. However, for this specific formula we can compute bounds and see $a_3 > 0$ by elementary means.
- A formal proof is given below; a plain-language summary follows.

Definition of a_3

The given code defines a *computable* real number via the sequence

$$a_3(n) = 1 + \left(\frac{1}{2}\right)^n - 72^{-n} \quad \text{for } n = 0, 1, 2, \dots$$

Equivalently,

$$a_3(n) = 1 + 2^{-n} - (72)^{-n}.$$

By definition, a_3 is the limit of this rational sequence:

$$a_3 = \lim_{n \rightarrow \infty} a_3(n).$$

Every $a_3(n)$ is a rational number, and the sequence is *computable*, so a_3 is a **computable real number**

².

Monotonicity and limits of the terms $(1/2)^n$ and 72^{-n}

For each $n \geq 0$, $(1/2)^n > 0$ and $72^{-n} > 0$. Since $0 < \frac{1}{2} < 1$, the sequence 2^{-n} is strictly decreasing:

$$2^{-0} = 1, \quad 2^{-1} = \frac{1}{2}, \quad 2^{-2} = \frac{1}{4}, \dots,$$

with $2^{-(n+1)} = 2^{-n}/2 < 2^{-n}$. By a standard limit fact for geometric sequences, $\lim_{n \rightarrow \infty} 2^{-n} = 0$. (Intuitively, each step halves the previous value, so it shrinks to 0.) Similarly, $0 < 1/72 < 1$ implies 72^{-n} is strictly decreasing:

$$72^0 = 1, \quad 72^{-1} \approx 0.0139, \quad 72^{-2} \approx 0.0001929, \dots,$$

and $\lim_{n \rightarrow \infty} 72^{-n} = 0$.

Thus both sequences $(1/2)^n$ and 72^{-n} **decrease to zero** as $n \rightarrow \infty$. In particular each term satisfies $0 \leq 2^{-n} < 1$ and $0 \leq 72^{-n} < 1$, and these terms tend to zero. (One can cite the general result: if $0 < r < 1$ then $\lim_{n \rightarrow \infty} r^n = 0$ ³.)

Behavior of the sequence $a_3(n)$

For each n ,

$$a_3(n) = 1 + 2^{-n} - 72^{-n}.$$

We analyze how $a_3(n)$ changes with n . At $n = 0$,

$$a_3(0) = 1 + 1 - 1 = 1.$$

For $n \geq 1$, note that $2^{-n} > 72^{-n}$ because

$$\frac{2^{-n}}{72^{-n}} = (72/2)^n = 36^n > 1 \quad (\text{for all } n \geq 1).$$

Therefore for $n \geq 1$,

$$a_3(n) = 1 + 2^{-n} - 72^{-n} > 1 + 2^{-n} - 2^{-n} = 1.$$

In fact, a direct check shows $a_3(1) = 1.486\dots$, and thereafter $a_3(n)$ decreases toward 1 (since each of the added terms 2^{-n} and subtracted 72^{-n} themselves decrease). One finds:

$$a_3 : 1.000, 1.4861, 1.2498, 1.1250, 1.06250, 1.03125, 1.015625, \dots$$

for $n = 0, 1, 2, 3, 4, 5, 6, \dots$ (see table). Formally, for $n \geq 1$ the difference

$$a_3(n+1) - a_3(n) = (2^{-(n+1)} - 2^{-n}) - (72^{-(n+1)} - 72^{-n}) = -2^{-n-1} + 72^{-n-1} \cdot (-71),$$

which is negative for $n \geq 1$ (since $36^{n+1} > 71$). Thus $a_3(n)$ is strictly decreasing for $n \geq 1$, but it increases from $n = 0$ to $n = 1$. Crucially, **every term** $a_3(n) \geq 1$, **and in fact** $a_3(n) > 1$ **for all** $n \geq 1$.

Since $a_3(n) \geq 1$ and $a_3(n) \rightarrow 1$ as $n \rightarrow \infty$, the limit is

$$a_3 = \lim_{n \rightarrow \infty} a_3(n) = 1 + 0 - 0 = 1.$$

Thus the real number a_3 defined by the code is exactly 1. It follows immediately that $a_3 > 0$. (More formally, the inequality $a_3(n) \geq 1$ for all n implies the limit satisfies $a_3 = \lim a_3(n) \geq 1$.)

Inequalities and asymptotics

To make the argument fully rigorous, one can use explicit inequalities. For $n \geq 1$:

$$0 < 72^{-n} < 2^{-n},$$

so

$$1 = 1 + 2^{-n} - 2^{-n} < 1 + 2^{-n} - 72^{-n} = a_3(n).$$

Thus $a_3(n) > 1$ for all $n \geq 1$. For $n = 0$ we have $a_3(0) = 1$. Hence $a_3(n) \geq 1$ for all n . As $n \rightarrow \infty$, both 2^{-n} and 72^{-n} tend to 0, so $a_3(n) \rightarrow 1$. In big-O terms,

$$a_3(n) = 1 + 2^{-n} - 72^{-n} = 1 + O(2^{-n}) \quad (n \rightarrow \infty),$$

since 72^{-n} is exponentially smaller.

The table below illustrates the first few values:

n	$a_3(n) = 1 + 2^{-n} - 72^{-n}$	Decimal approx.
0	$1 + 1 - 1 = 1$	1.0000000000
1	$1 + 1/2 - 1/72 = 107/72$	1.4861111111
2	$1 + 1/4 - 1/72^2 = 6479/5184$	1.2498070988
3	$1 + 1/8 - 1/72^3 = 419903/373248$	1.1249973208
4	$1 + 1/16 - 1/72^4 = 28553471/26873856$	1.0624999628
5	$1 + 1/32 - 1/72^5 = 1995383807/1934917632$	1.0312499995
6	$1 + 1/64 - 1/72^6 = 141490851839/139314069504$	1.01562499999

These values confirm $a_3(n) \geq 1.0156$ for $n \leq 6$, approaching 1 from above.

Computability and decidability of positivity

By construction, a_3 is *computable*: the code gives an explicit formula for each term. However, a key point in computable analysis is that **no algorithm can decide the sign (positive/zero/negative) of an arbitrary computable real** from its code. In fact, it is a known theorem that the ordering of computable reals is not decidable ¹. Wikipedia explains:

“The order relation on the computable numbers is not computable... there is no Turing machine which on input of a machine description for a real number a outputs ‘YES’ if $a > 0$ and ‘NO’ if $a \leq 0$.” ¹.

The usual proof reduces from the Halting problem: one constructs a computable real that is positive if and only if a given Turing machine halts ¹. Thus, in **general**, a uniform algorithm to decide positivity of an unknown computable real does not exist.

In our concrete case, however, we have a **special closed-form sequence**. By elementary algebra we have shown $a_3(n) \geq 1$ for all n , so the limit $a_3 \geq 1$. Thus we **conclude** $a_3 > 0$ with certainty. This is a decision made by analysis of the formula, not by a general algorithm. It illustrates that *specific* computable reals can often be resolved by hand, but no single program works for all.

Formal proof

Theorem. Let $a_3 = \lim_{n \rightarrow \infty} (1 + 2^{-n} - 72^{-n})$. Then $a_3 > 0$. In fact $a_3 = 1$.

Proof. For $n = 0$ we have $a_3(0) = 1 + 1 - 1 = 1$. For $n \geq 1$, since $72^n = (36 \cdot 2)^n = 36^n \cdot 2^n$ and $36^n > 1$, we have

$$2^{-n} > 72^{-n}.$$

Hence for each $n \geq 1$,

$$a_3(n) = 1 + 2^{-n} - 72^{-n} > 1 + 2^{-n} - 2^{-n} = 1.$$

Thus $a_3(n) \geq 1$ for all n (with equality only at $n = 0$). Taking limits as $n \rightarrow \infty$, the two small terms 2^{-n} and 72^{-n} both tend to 0, so

$$a_3 = \lim_{n \rightarrow \infty} a_3(n) = 1 + 0 - 0 = 1.$$

In particular $a_3 = 1 > 0$. This completes the proof. (Monotonicity arguments above ensure the limit exists and equals 1.) ■

Plain-language summary

The program's code builds a sequence of numbers $a_3(n)$ and defines a_3 as the number it converges to. Each term is $1 + (1/2)^n - (1/72)^n$. As n grows, both $(1/2)^n$ and $(1/72)^n$ shrink rapidly to 0. One checks that even though $a_3(1)$ is about 1.486, the sequence then decreases toward 1 and never drops below 1. Indeed every term is at least 1. Therefore the limit is 1, and clearly 1 is positive. So the real number defined by the code is positive.

In general, figuring out whether a computable real given by a program is positive can be as hard as solving the halting problem ¹. But here, thanks to the simple form $1 + 2^{-n} - 72^{-n}$, we can decide by straightforward algebra. Our rigorous analysis above shows $a_3(n) \geq 1$ for all n , so the limit is 1; hence $a_3 = 1 > 0$.

¹ ² Computable number - Wikipedia
https://en.wikipedia.org/wiki/Computable_number

³ Geometric series - Wikipedia
https://en.wikipedia.org/wiki/Geometric_series